The Questions

1.

a)

i) What are the differences between the lightning and switching transients? [3]

Ans: Lightning transients result from sudden discharge from the electrostatically charged clouds in the air. This is related to weather. The switching transients result from operation of circuit breaker opening/closing power system components such as transformer, generator, lines. These operations can be deliberate or can happens when protection system operates.

[3]

ii) How are these transients electrically characterised? [4]

Ans : lightning transient is represented as unidirectional current in kA that rises fast and decays slowly . Typically, 1.2 microseconds rise time to max amps and then 50 microseconds. The current ranges in the tens of kAs. [2]

Switching surge on the other hand is represented as large over voltages such as 500kv in a typical 275kv system. It lasts for hundreds of micro seconds. The models are useful for insulation design and co-ordination

[2]

b)

i) What is *resonance stability* in the context of power system operation? [5]

Ans: The term resonance stability encompasses subsynchronous resonance (SSR), whether it be associated with an electromechanical resonance or an entirely electrical resonance. The term SSR can manifest in two possible forms: (i) due to a resonance between series compensation and the mechanical torsional frequencies of the turbine-generator shaft, and (ii) due to a resonance between series compensation and the electrical characteristics of the generator. The first of these occurs between the series compensated electrical network and the mechanical modes of torsional oscillations on the turbine-generator shaft, while the second is a purely electrical resonance and termed Induction Generator Effect (IGE). Hence, the resonance stability has been split into two categories: mechanical and electrical.

[5]

ii) How has the nature of the resonance stability been influenced by the increasing presence of high-power electronics interfaced generation in the power network? [3]

There are now increasing possibilities of electrical resonances between network components. The converter control couple of with phase lock loop in grid following converter have added new dimension to these electrical resonances. The frequency is now wide ranging with in sub synchronous range. Wind farm also have exhibited SSR through IGE. FACTS devices with new control reported to have sub synchronous torsional as well as control interactions.

[3]

iii) Why is the electromagnetic transient model required for resonance stability study whereas RMS phasor model is adequate for electromechanical stability study? [4]

Ans: Resonance stability is both electrical and electromechanical in nature. In both categories the frequency of oscillations ranges from few Hz to tens of Hz. It is required to have electromagnetic models of the electrical networks to study the response in this frequency range.

[2]

Traditional electromechanical stability such as small signal stability to analyse local, and inter-area modes are restricted to max 2.5 Hz. So, power frequency transients in electrical circuit is not needed for the analysis. the RMS phasors are good enough to capture the behaviours in time domain following events leading to their manifestation.

[2]

iv) Describe the purpose of a power system stabiliser (PSS) in an interconnected power system.

[4]

Small signal oscillatory instability is the consequence of high gain fast

acting excitation system. While fast excitation system guarantees the transient stability of a large fossil fuelled unit, the post fault equilibrium can be oscillatory stable or even unstable. The negative damping torque introduced by the AVR response needs counter action through additional damping signal. This comes in the form of power system stabiliser (PSS) which should only act when there is oscillations and filter out any very low frequency (less than 0.1 Hz) variation of the stabilising signal. Without PSS stable operation of the synchronous generator utilising its full capacity is not possible.

[4]

ii) List two stabilising feedback signals that can be used in PSS for providing stability in system operation.

[2]

Ans: (i) rotor speed, (ii) output electrical power

[25 marks]

2.

a)

i) The rotor of a synchronous generator is of solid construction whereas the stator is made up of laminated materials. Provide technical reasons for such a constructional practice. [4]

The students are expected to say stator carries power frequency currents – so losses due to hysteresis and eddy currents in the magnetic materials are minimised

[2]

In the rotor it is DC current – so the relative velocity between rotor and rotor winding is zero – no induced current – similarly impact of stator MMF on the rotor body at synchronous speed is nil – so rotor does not need to be laminated

[2]

ii) Even though the rotor of a synchronous generator is of solid construction, it is reinforced with additional windings. Describe the purpose of such windings. [4]

The students are expected to demonstrate their understanding of interaction between rotor and stator winding during transient situation i.e non- synchronous speed.

[2]

One would expect that the frequency of the current is the difference between synchronous frequency and rotor speed. The usefulness of additional winding to provide counter toque to bring the rotor back to initial operating speed.

[2]

iii) How are these additional rotor windings modelled for stability study?

[2]

These are closed circuit winding. They are represented through equivalent resistance, self-inductance and mutual inductance with stator. So one first order differential winding per circuit with flux as state variable along with open circuit time constant as function of resistance and inductance is used to represent it.

[2]

(iv) Why do we ignore the presence/contribution of these windings in steady state analysis of synchronous generator?

[2]

Ans: at steady state the winding spins at synchronous speed. They do not have any electromagnetic/mechanical impact on the stator output voltage

[2]

iii) Modern synchronous generators are designed to have a large steady state synchronous reactance (between 1.5 to 2.0 pu). However, when a direct terminal short circuit occurs, the initial currents are 4-5 times higher than normal load current. Explain the reason. [5]

The students should demonstrate their understanding that during fault synchronous reactance does not exist

[2]

They then mention about the electromagnetic flux linkage geometry during fault – sub transients, transients etc and how the varying flux path lead to variable inductance with time .. how that lead to larger currents etc.

[3]

iv) Why does a synchronous generator offer higher overload limit while overexcited compared to when it is under excited?

[4]

The flux reinforcement during the leading power factor based synchronous generation saturates the end ring and increases the eddy current.

[2]

This leads to the losses and heating in the end winding region – thus the permitted output is less than that as lagging power operation.

[2]

v) What is saliency power in the contexts of synchronous machine operation?

[2]

Because of the variation of the gap between stator and rotor, electromagnetic reluctance torque is exerted by the stator which is balanced by the rotor through mechanical balancing torque. This results in steady production of power.

[2]

vi) How does it vary with generator load angle (δ)?

[2]

It is a sinusoidal function of twice the load angle.

[2]

[25 marks]

3.

a)

i) What is Park Transformation (d-q) in the contexts synchronous machine dynamics?

[3]

I am expecting the students to state that the magnetic effect of stator can be characterised by two windings in quadrature and when those two windings are aligned on the polar and inter-polar axis – with the same reference as rotor – the inductances are fixed – easy to analyse.

[3]

ii) Why is Park Transformation preferred over a stationary 3-phase (a-b-c) reference frame?

[3]

Stator self and mutual inductance in phase domain is function of rotor angle. It is dictated by the loading. So, one needs to express inductance which remains fixed – in a-b-c frame it is not possible.

[3]

iii) Despite having represented a synchronous machine in d-q reference frame, why another transformation is required when the generator under analysis is part of an interconnected power system network? [3]

Each machine is modelled in its own dq (Park) co-ordinate taking the terminal voltage as reference– the load angle of each machine is different from the other machine – so with respect to the network or Kron reference frame each individual machine must be represented again requiring machine (Park) to Network (Kron) transformation.

[3]

iv) Why is it difficult to implement fast control for hydro turbine? [3]

The transfer function of a hydro turbine has a right-half plane zero. This introduces 90 degrees phase lag at high frequency. So, for very fast governor control output power is not produced immediately – rather phase margin may reach negative leading to instability. The situation is addressed through transient gain reduction (TGR) block in the governor control loop.

[3]

v) Why is stator transient neglected in transient stability study? [3]

Transient stability involves dynamic response of large rotor mass which is governed by the swing equation driven by the balance between the mechanical and electrical torque. Because of the inertia, it takes about a second for the rotor angle to reach the first swing. The stator transient is

caused by power frequency which introduces transformer EMF in each axis and is very negligible compared to rotational EMF. Because of this reason, they are neglected. Their inclusion introduces a power frequency modulation in the rotor swing response and usually produces less swing angle than when their effects are included. The transient stability margin estimated this way produces a conservative value. The advantage is analysis is simple as the model is.

[3]

iv) A synchronous generator produces a voltage of 20 kV line to line at the generator terminal bus. This bus is part of a large network which has a reference node at another location. The power flow solution produces an angle (θ) of 5 degrees for this bus voltage. The generator load angle (δ) is 35 degrees. Compute the direct and quadrature axis components of this terminal voltage. [4]

The students would be able to apply the formula in part (iii) with appropriate choice of angle. in this case Vq+jVd = 20*exp(-30 degree)

Vq = 20.0*cos(-30) , Vd =-20*sin(-30): Ans Vq=17.32 kV and Vd = -10.0 kV

[4]

b)

i) Despite having no power electronics and cheaper cost why are Type 1 and Type 2 wind turbine technologies not preferred in modern wind farms? [3]

Type 1 only operates at more than synchronous speed which is not allowed to vary. Type 2 has rotor resistance assisted speed variation but caters for a small speed range. Both requires reactive power supports. In widely varying wind speed situation, the power is not captured very efficiently.

[3]

ii) A wind turbine is designed with 4 m/s cut-in speed and 24 m/s cut-out speed. About 400 such turbines are installed in a large offshore wind farm. The wind hits the first row with a speed of 4.2 m/s. It was observed that some of the turbines in the third row counted from the incoming wind direction were not spinning. Explain clearly what could have been the reason(s). [3]

This requires the understanding of wake effect in wind farm – slowing down of wind speed by the turbine up stream – this means wind after heating the first row – will slow down – the speed being very close to cut-in speed – after wake effect it will be below cut-in value for the third row, so some turbine may not be hit by cut-in speed.

[3]

[25 marks]

4.

a)

i) Why are gearboxes necessary in all of first three types of wind turbine generator? [2]

Generally, wind turbine rotates at 4-6 RPM depending on the wind speed/radius of the blade. This is way too low to generate electrical power at power frequency such as 50 and or 60 Hz. The shaft to which the generator is connected rotates at high speed. The high shaft speed is produced by the gear box – having lesser teeth on the generator shaft and larger teeth on the turbine side shaft.

[2]

iii) How does one model the gear box for stability studies? [2]

Usually gear box is modelled by a scalar gain. When speed is expressed in per unit, it can even be simplified as unity gain. The shaft on either side can be modelled by spring mass which can be referred to either side of the gear box through the square of the gear box ratio.

[2]

iv) Why does Type 4 wind turbine usually have large number of poles? [3]

I expect the student to write along the line like this It is required to have reasonably good frequency of power generation at the stator terminal for optimising magnetics design – in the absence of gear box, large number of poles will produce that frequency at lower wind turbine speed … etc..

[3]

b)

i) How does the *stall* differ from the *pitch* in the contexts of wind power generation control? [3]

Pitching and stalling are both mechanisms used to control the torque exerted on a wind turbine blade and hence control power production. Pitch is an angle between chord line and the direction of motion of the blade. The angle between the relative direction of wind and the chord-line is the angle of attack. By controlling the pitch angle slightly, the angle of attack is adjusted. The blades avoid part of the wind and power output is controlled. Usually lift force is high, drag force is also low. When excessive pitching is done in the negative angle range the angle of attack is very high. The drag forces increase significantly, lift forces drop drastically. This is known as stall. Stalling is avoided as it causes sheer, vibration and fatigue on the blade as well as stressing the pitching bearing structure. It is always recommendable to have an optimal angle of attack by pitch and yaw control show that the lift to drag force ratio is largest for operating the turbine at maximum efficiency over its 20-30 years life.

[3]

ii) How is stall control implemented? [3]

The stall control is implemented in two ways: passive stall control and active stall control. In passive stall control, the aero foil geometry of the blade is shaped in such a way that the lift forces are automatically dropped at a wind speed higher than the rated speed. The turbine speed is dropped, torque and power produced stays at rated or even drops at more than rated wind speed. In active stall control, blade angle is adjusted through pitching mechanism to create high angle of attack beyond optimal value, thus drag forces increase and lift forces decrease. This slows down the turbine speed as well caps the power at rated value when wind speed is excessively higher than rated speed.

iii) Why is the stall control mechanism avoided in all modern wind turbines? [3]

For small size turbine passive stall control is fine. With large modern turbines the stall control involves large twist of blade around the bearing. Such action leads to sheer on the blade and so blade fatigues and stress on the bearing structure. Wind turbines are designed to last for about 25 years. Any excessive sheering stress on blade mounting structure reduces the operating life of the turbines. This is the reason stall is avoided in modern design.

[3]

iv) Why is the generator torque/power control preferred over the pitch control at the sub-rated wind condition? [3]

I expect the student to mention important aspect about mechanical control – it is designed to have at higher than rated speed because these controls always add operating stress and fatigue on the components. Usually pitch angle is kept at minimum = 0.0, and any change from this result in drop in torque and hence reduced power production. At sub-rated condition, this is not desirable – so torque is increased by increasing current through rotor control to optimally capture the available power – target is to operate at optimal tip-speed ratio- this can be done through control from the electrical side.

[3]

c)

(i) Why is the *transient stability* less of a concern to the system operators with significant wind generation?

[3]

I am expecting the student to demonstrate their understanding that transient stability is more of a problem associated with large synchronous machine with huge mass and electromechanical coupling. Generally, wind generators are asynchronously connected to the network through power electronic interface. The control in power converter limit the current fed to the network by the WTG during disturbance. WTG on its own is not part of large synchronous system. Any terminal fault is managed through well designed fault ride through capability which are part of grid connection requirement. So transient stability is not a big concern in a system with large proportion of wind generations.

[3]

(ii) How can wind generators be operated to provide the *frequency stability* in power system operation? [3]

My expectation from students is they will state the capability of a WTG for fast as well as slow primary frequency control through inertial control (slowing down the turbine to extract kinetic energy) temporarily. Then for slow primary, they expect to state about droop control and operating the WTG at less than MPPT point etc.

[3]

[25 marks]